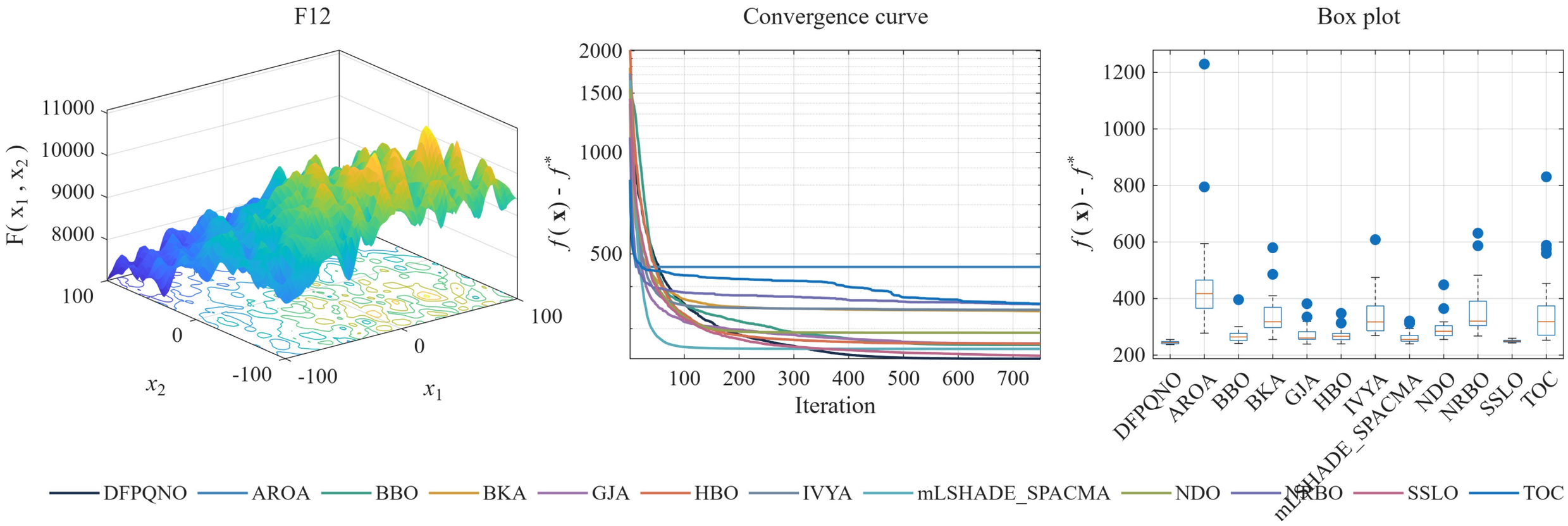

F12
F( x1 , x2 )
11000
10000
9000
8000
100
0
-100
x2
-100
0
100
x1
Convergence curve
f( x) - f*
2000
1500
1000
500
100
200
300
400
500
600
700
Iteration
Box plot
f( x) - f*
1200
1000
800
600
400
200
DFPQNO
AROA
BBO
BKA
GJA
HBO
IVYA
mLSHADE_SPACMA
NDO
NRBO
SSLO
TOC
DFPQNO
AROA
BBO
BKA
GJA
HBO
IVYA
mLSHADE_SPACMA
NDO
NRBO
SSLO
TOC